

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			$n = \frac{pV}{RT}$ (1) $n = \frac{1.01 \times 10^5 \times 39 \times 10^{-6}}{8.31 \times 360}$ (1) $n = 1.317 \times 10^{-3} \text{ mol}$ (1) $M_r = \frac{0.181}{1.317 \times 10^{-3}} = 137.4$ $M_r \text{ PCl}_3 = 137.5$ therefore chloride is PCl_3 (1) accept alternative method e.g. assume PCl_3 and work back to show volume of 39 cm^3	1	1				
	(e)	(i)		$(\text{P}^{35}\text{Cl}_2)^+$ accept if charge missing			1	1		
		(ii)		chlorine has isotopes ^{35}Cl and ^{37}Cl in ratio of 3:1 / 75% to 25% (1) since peak C due to $(\text{P}^{35}\text{Cl}_2)^+$ and peak E due to $(\text{P}^{37}\text{Cl}_2)^+$ height of C : E is $(3 : 1) \times (3 : 1)$ i.e 9 : 1 (1)		1	1	2		
				Question 11 total	1	10	6	17	7	0